

ECE 461/661 Spring 2025 Mini-Exam 1

Introduction to Machine Learning for Engineers

Prof. Gauri Joshi and Prof. Carlee Joe-Wong

Mon, February 9, 2025, 9:00am - 9:40am PT/12:00pm - 12:40pm ET/7:00pm - 7:40pm CAT

Instructions

- If a problem asks you which of its choices is TRUE, you should treat choices that may be either true or false as FALSE.
- Unless otherwise stated, only one option is correct in each multiple-choice question.** No partial credit will be given for multiple-choice or true/false questions.
- For descriptive questions, make sure to explain your answers and reasoning. We will give partial credit for wrong answers if portions of your reasoning are correct. Conversely, correct answers accompanied by incomplete or incorrect explanations may not receive full credit.
- You are allowed one **physical, single-sided** US-letter or A4 sized cheat sheet. No other notes or material (aside from blank pieces of scratch paper) may be used.
- You may only use a pen/pencil, eraser, and scratch paper. The backside of each sheet in the exam can also be used as scratch paper. If you do not wish for us to grade your scratch work, please clearly indicate which parts of your work we should ignore.
- Calculators are permitted but not necessary. If you choose to use a calculator, it must be a standalone one. No other electronic devices such as phones, tablets or laptops can be used during the exam.
- If you would like to ask a clarification question during the exam, raise your hand and an instructor or TA will come over. Note that we will not help you answer the questions but can give clarifications.

Problem	Type	Points
1-5	True/False	7
6-8	Multiple Choice	6
9	Descriptive	4
10	Descriptive	3
11	Honor Pledge	0
Total		20

1 True or False (7 points)

Problem 1: [1 points] The eigenvalues of $\mathbf{A}^\top \mathbf{A}$ or $\mathbf{A}\mathbf{A}^\top$ are squares of the singular values of matrix $\mathbf{A}$.

- ☐ True
☐ False

Problem 2: [1 points] Suppose you wish to estimate the probabilities of rolling each side of a given six-sided die, based on 100 observed outcomes from rolling it. You may assume the 100 rolls are independent. Based on your knowledge of the manufacturing process for this die, you also have a prior distribution π for the probabilities of rolling each side. Using MAP with the prior distribution π will produce a distribution more similar to the true distribution than using MLE.

- ☐ True
☐ False

Problem 3: [1 points] Suppose you flipped a coin twice, and that the first time it landed on HEADS and the second time on TAILS. You model the HEADS outcome as having a Bernoulli distribution with parameter π . Then the log-likelihood of your observed data is $\log(\pi) + \log(1 - \pi)$.

- ☐ True
☐ False

Problem 4: [1 points] Given a feature matrix $\mathbf{X}$ and label vector $\mathbf{y}$, linear regression finds a weight vector $\mathbf{w}^*$ such that $\mathbf{X}\mathbf{w}^* = \mathbf{y}$.

- ☐ True
☐ False

Problem 5: Consider a regression problem with a feature vector $\mathbf{x} = [x_1, x_2, x_3] \in \mathbb{R}^3$ and target $y \in \mathbb{R}$, where we find the model $\mathbf{w}$ that minimizes the residual sum of squares function (without adding any regularizer). We consider two different feature transformations of $\mathbf{x}$: $\phi_A = [x_1, x_2, x_3, x_1^2, x_2^2, x_3^2]$, $\phi_B = [\frac{x_1+x_3}{2}, \frac{x_1+x_2}{2}, \frac{x_2+x_3}{2}]$. Indicate whether the following statements are true or false:

- a. [1 points] The model $\mathbf{w}_A$ learnt using ϕ_A has higher bias and lower variance than the model $\mathbf{w}_B$ learnt using ϕ_B .
☐ True
☐ False
- b. [1 points] The model $\mathbf{w}_B$ learnt using ϕ_B has higher bias and lower variance than the model $\mathbf{w}$ learnt using the original feature vector $\mathbf{x}$.
☐ True
☐ False
- c. [1 points] The model $\mathbf{w}_B$ learnt using ϕ_B is identical to the model $\mathbf{w}$ learnt using the original feature vector $\mathbf{x}$, that is, $\mathbf{w}_B = \mathbf{w}$.
☐ True
☐ False

2 Multiple Choice (6 points)

Problem 6: [2 points] If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of a matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, what are the eigenvalues of the matrix $\mathbf{A}^2$?

- ☐ $\lambda_1, \dots, \lambda_n$
- ☐ $2\lambda_1, \dots, 2\lambda_n$
- ☐ $\lambda_1^2, \dots, \lambda_n^2$
- ☐ None of the above

Problem 7: [2 points] Recall that ridge regression learns a parameter vector $\mathbf{w}$ given a feature matrix $\mathbf{X}$ and label vector $\mathbf{y}$. Which of the following statements is/are true about ridge regression? **More than one option can be true.**

- ☐ Suppose that our data has d features. For sufficiently large d , ridge regression ensures that some entries in $\mathbf{w}$ will be equal to 0.
- ☐ Ridge regression adds an ℓ_2 regularization term to the standard linear regression formulation.
- ☐ Ridge regression worsens the model's generalization to unseen data, compared to the model obtained with linear regression.
- ☐ Ridge regression helps limit the magnitude of the entries of $\mathbf{w}$.

Problem 8: [2 points] Which of the following best describes the rank of a matrix?

- ☐ The rank is equal to the number of rows in the matrix.
- ☐ The rank is equal to the number of columns in the matrix.
- ☐ The rank is the maximum number of linearly independent rows (or columns) of the matrix.
- ☐ A matrix is invertible if and only if its rank is zero.

3 Descriptive (7 pts)

Problem 9: [4 points] Consider the dataset given by $\left\{ \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, y_1 \right), \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}, y_2 \right) \right\}$. You decide to apply linear regression to this dataset, i.e., to solve the following optimization problem:

$$\min_{w_1, w_2} \left\| \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} - \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \right\|_2^2. \quad (1)$$

- Find the values of w_1 and w_2 that solve the linear regression problem in Equation (1), and find the resulting squared ℓ_2 loss of this model. Your answer may be in terms of y_1 and y_2 .
- You now decide to compare your model from part (a) to the solution of a ridge regression model with regularization parameter λ . In other words, your new parameters w_1, w_2 will solve the optimization problem

$$\min_{w_1, w_2} \left\| \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} - \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \right\|_2^2 + \lambda (w_1^2 + w_2^2). \quad (2)$$

Find a closed-form solution for w_1, w_2 that solves Equation (2). Your answer may be in terms of y_1, y_2, λ .

Does the squared ℓ_2 loss

$$\left(\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} - y_1 \right)^2 + \left(\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} - y_2 \right)^2$$

increase or decrease compared to the loss value you found in part (a)?

- Now suppose that a new data point, $\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, 1 \right)$ has arrived. You decide to re-solve for the linear regression model with all three of your data points $\left\{ \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, y_1 \right), \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}, y_2 \right), \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, 1 \right) \right\}$, i.e., you wish to solve

$$\min_{w_1, w_2} \left\| \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} - \begin{bmatrix} y_1 \\ y_2 \\ 1 \end{bmatrix} \right\|_2^2. \quad (3)$$

Find a closed-form solution for w_1, w_2 that solves Equation (3). Your answer may be in terms of y_1, y_2 .

- Suggest a way to modify the linear regression problem from part (c) that will allow you to reduce the ℓ_2 loss on your three data points $\left\{ \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, y_1 \right), \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}, y_2 \right), \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, 1 \right) \right\}$. Your answer should be 1-2 sentences long.

WRITE YOUR ANSWER TO PROBLEM 9 BELOW

CONTINUE YOUR ANSWER TO PROBLEM 9, IF NEEDED

Problem 10: *[3 points]*

Suppose that the time X taken by a generative AI model to respond to a prompt follows an exponential distribution with rate μ , where the probability density function of the exponential distribution is given by

$$f_X(x) = \lambda e^{-\lambda x} \quad \text{for all } x \geq 0$$

For 5 prompts, we observe that the response times are 1, 0.5, 2, 1.25 and 0.25 respectively.

- a. What is the maximum likelihood estimate of λ ?
- b. Suppose we put a prior on λ whose probability density lies in the range $[0, 1]$ and is concentrated around 0.5, and we find the MAP estimate λ_{MAP} . Is λ_{MAP} greater or less than the MLE estimate λ_{MLE} that you found in part (a)? Explain your reasoning in 1-2 sentences.

WRITE YOUR ANSWER TO PROBLEM 10 BELOW

CONTINUE YOUR ANSWER TO PROBLEM 10, IF NEEDED

4 Honor Pledge

Problem 11: *[0 points]* To affirm that you did not cheat on the exam, please write out the below statement. Sign your name beneath it. **Failure to do so will be taken as a sign that you have cheated on the exam.**

I pledge my honor that I neither gave nor received unauthorized assistance on this examination.